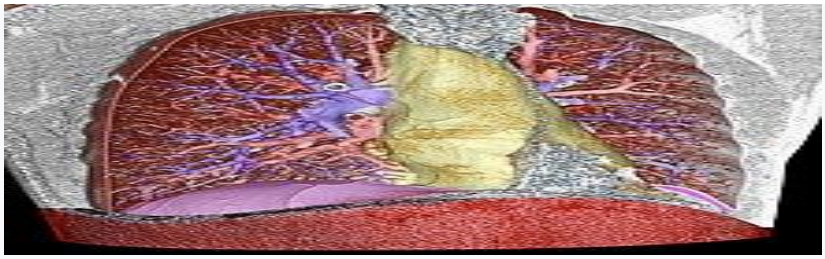
Unit-5 (Image Segmentation & Representation)Point Detection

Lecture-1

In digital image processing and computer vision, **image segmentation** is the process of partitioning a digital image into multiple **image segments**, also known as **image regions** or **image objects** (sets of pixels). The goal of segmentation is to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze.[1][2] Image segmentation is typically used to locate objects and boundaries (lines, curves, etc.) in images. More precisely, image segmentation is the process of assigning a label to every pixel in an image such that pixels with the same label share certain characteristics.

The result of image segmentation is a set of segments that collectively cover the entire image, or a set of contours extracted from the image (see edge detection). Each of the pixels in a region are similar with respect to some characteristic or computed property , such as color, intensity, or texture. Adjacent regions are significantly different color respect to the same characteristic(s). When applied to a stack of images, typical in medical imaging, the resulting contours after image segmentation can be used to create 3D reconstructions with the help of interpolation algorithms like marching cubes.[4]

Applications[edit]



Volume segmentation of a 3D-rendered CT scan of the thorax: The anterior thoracic wall, the airways and the pulmonary vessels anterior to the root of the lung have been digitally removed in order to visualize thoracic contents:

– blue: pulmonary arteries
– red: pulmonary veins (and also the abdominal wall)
– yellow: the mediastinum
– violet: the diaphragm

Some of the practical applications of image segmentation are:

- Content-based image retrieval
- Machine vision
- Medical imaging, including volume rendered images from computed tomography and magnetic resonance imaging.
  - Locate tumors and other pathologies

    - Measure tissue volumes
    - Diagnosis, study of anatomical structure[Surgery planning
    - Virtual surgery simulation
    - Intra-surgery navigation
    - Radiotherapy[
- Object detection
    - Pedestrian detection
    - Face detection
    - Brake light detection
    - Locate objects in satellite images (roads, forests, crops, etc.)
- Recognition Tasks
    - Face recognition
    - Fingerprint recognition
    - Iris recognition
- Traffic control systems
- Video surveillance
- Video object co-segmentation and action localization

Several general-purpose algorithms and techniques have been developed for image segmentation. To be useful, these techniques must typically be combined with a domain's specific knowledge in order to effectively solve the domain's segmentation problems.

Classes of segmentation techniques

There are two classes of segmentation techniques.

- Classical computer vision approaches
- AI based techniques

Groups of image segmentation[edit]

- **Semantic segmentation** is an approach detecting, for every pixel, belonging class of the object For example, when all people in a figure are segmented as one object and background as one object.
- **Instance segmentation** is an approach that identifies, for every pixel, a belonging instance of the object. It detects each distinct object of interest in the image. For example, when each person in a figure is segmented as an individual object.
- **Panoptic segmentation** combines semantic and instance segmentation. Like semantic segmentation, panoptic segmentation is an approach that identifies, for every pixel, the belonging class. Unlike semantic segmentation, panoptic segmentation distinguishes different instances of the same class.

Thresholding

The simplest method of image segmentation is called the thresholding method. This method is based on a clip-level (or a threshold value) to turn a gray-scale image into a binary image.

The key of this method is to select the threshold value (or values when multiple-levels are selected). Several popular methods are used in industry including the maximum entropy method, balanced histogram thresholding, Otsu's method (maximum variance), and k-means clustering.

Recently, methods have been developed for thresholding computed tomography (CT) images. The key idea is that, unlike Otsu's method, the thresholds are derived from the radiographs instead of the (reconstructed) image.

New methods suggested the usage of multi-dimensional fuzzy rule-based non-linear thresholds. In these works decision over each pixel's membership to a segment is based on multi-dimensional rules derived from fuzzy logic and evolutionary algorithms based on image lighting environment and application.[1]

Now, in this mask processing operation, shift this mask over the entire image to calculate some weighted sum of pixel at a particular location.

Say for example, if I place this mask at a location (x, y) in our original image; then using allother different mask coefficients, calculate the value R. Use of this mask we have seen in connection with image sharpening while we have taken different values of the coefficients. In case of image smoothing, we have taken the values of the mask coefficients to be all ones. So, that leads to an image averaging operation. So, depending upon what are the coefficient values ofthis mask that we choose, we can have different types of image processing operations.

Now here, you find that when I use this mask, then depending upon the nature of the image around point (x, y), I will have different values of R

[1] Point Detection

Point Detection

- This is used to detect isolated spots in an image.
- The graylevel of an isolated point will be very different from itsneighbors.
- It can be accomplished using the following 3□3 mask:

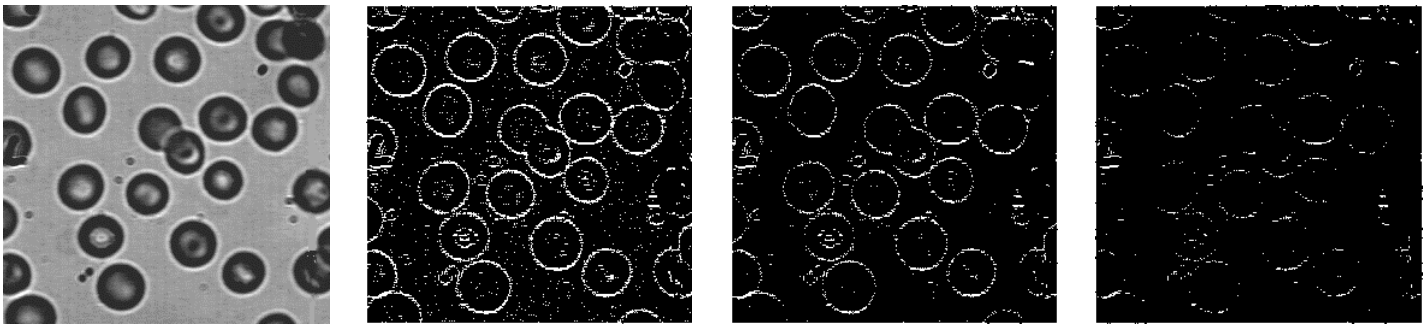
□1	□1	□1
□1	8	□1
□1	□1	□1

- The output of the mask operation is usually thresholded.

- We say that an isolated point has been detected if

$$8f_5 - (f_1 + f_2 + f_3 + f_4 + f_6 + f_7 + f_8 + f_9) \geq T$$

for some pre-specified non-negative threshold T.



Orig. Image $T = 0.25$ $T = 0.35$ $T = 0.5$

Unit-5 (Image Segmentation & Representation)

Line Detection

Lecture-2

Discontinuity based Image Segmentation

[1] Points

[2] Lines

[3] Edges

Point Detection

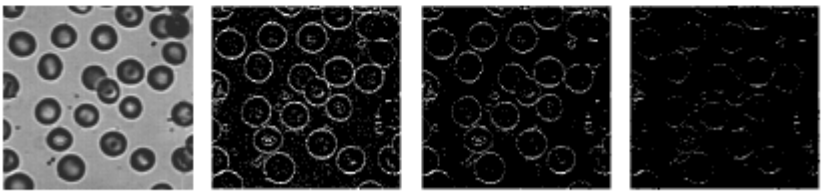
- This is used to detect isolated spots in an image.
- The graylevel of an isolated point will be very different from its neighbors.
- It can be accomplished using the following 3×3 mask:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

- The output of the mask operation is usually thresholded.
- We say that an isolated point has been detected if

$$|8f_5 - (f_1 + f_2 + f_3 + f_4 + f_6 + f_7 + f_8 + f_9)| > T$$

for some pre-specified non-negative threshold T.



Orig. Image $T = 0.25$ $T = 0.35$ $T = 0.5$

Line Detection -Line detection is an important step in image processing and analysis. Lines and edges are features in any scene, from simple indoor scenes to noisy terrain images taken by satellite. Most of the earlier methods for detecting lines were based on pattern matching. The patterns directly followed from the definition of a line. These pattern templates are designed with suitable coefficients and are applied at each point in an image. A set of such templates is shown in If the first mask were moved around an image, it would respond more strongly to lines oriented horizontally. With constant background, the maximum response would result when the line passed through the middle row of the mask. This is easily verified by sketching a simple array of 1's with a line of a different gray level running horizontally through the array. A similar experiment would reveal that the second mask in responds best to lines oriented at +45; the third mask to vertical lines; and the fourth mask to lines in the – 45 direction. These directions can also be established by noting that the preferred direction of each mask is weighted with larger coefficient i.e., 2 than other possible directions.

Let R1, R2, R3 and R4 denote the responses of the masks in from left to right, where the R's are given by equation 9.2. Suppose that all masks are run through an image. If, at a certain point in the image, for all j ≠ I, that point is said to be more likely associated with a line in the direction of mask i. For example, if at a

point in the image, for j = 2, 3, 4, that particular point is said to be more likely associated with a horizontal line.

Detection of lines

- This is used to detect lines in an image.
- It can be done using the following four masks:

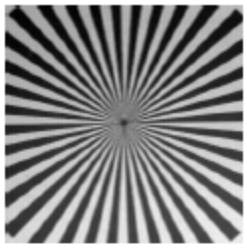
$$\mathcal{D}_{0^\circ} = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$ Detects horizontal lines

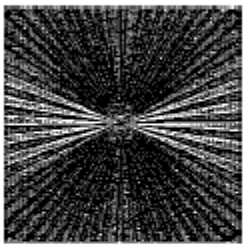
$$\mathcal{D}_{45^\circ} = \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$ Detects 45° lines

$$\mathcal{D}_{90^\circ} = \begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$ Detects vertical lines

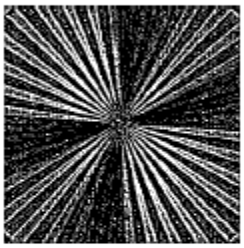
$$\mathcal{D}_{135^\circ} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$ Detects 135° lines

- Let R_{0°, R_{45°, R_{90°, and R_{135°, respectively be the response to masks $\mathcal{D}_{0^\circ}$, $\mathcal{D}_{45^\circ}$, $\mathcal{D}_{90^\circ}$, and $\mathcal{D}_{135^\circ}$, respectively. At a given pixel (m,n), if R_{135° is the maximum among $\{R_{0^\circ}, R_{45^\circ}, R_{90^\circ}, R_{135^\circ}\}$, we say that a 135° line is most likely passing through that pixel.





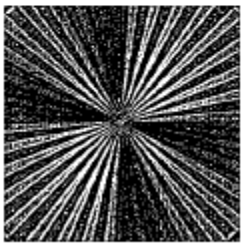
$R_{0^\circ} = \max\{R_{0^\circ}, R_{45^\circ}, R_{90^\circ}, R_{135^\circ}\}$



$R_{45^\circ} = \max\{R_{0^\circ}, R_{45^\circ}, R_{90^\circ}, R_{135^\circ}\}$



$R_{90^\circ} = \max\{R_{0^\circ}, R_{45^\circ}, R_{90^\circ}, R_{135^\circ}\}$



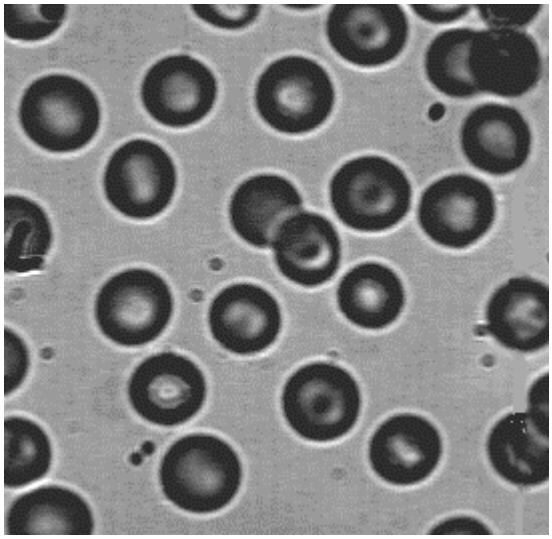
$R_{135^\circ} = \max\{R_{0^\circ}, R_{45^\circ}, R_{90^\circ}, R_{135^\circ}\}$

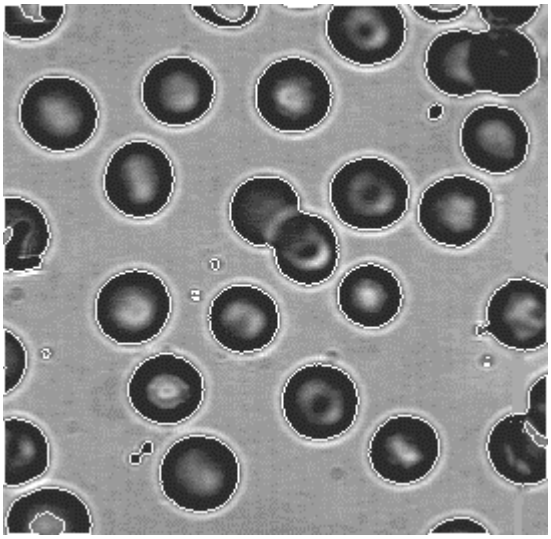
rtical lines

Image Segmentation

- Segmentation is an image analysis task that subdivides an image into disjoint regions of interest for further analysis. It is usually thefirst step in **image analysis**.

- The disjoint regions usually correspond to the different objects of interest in the image.

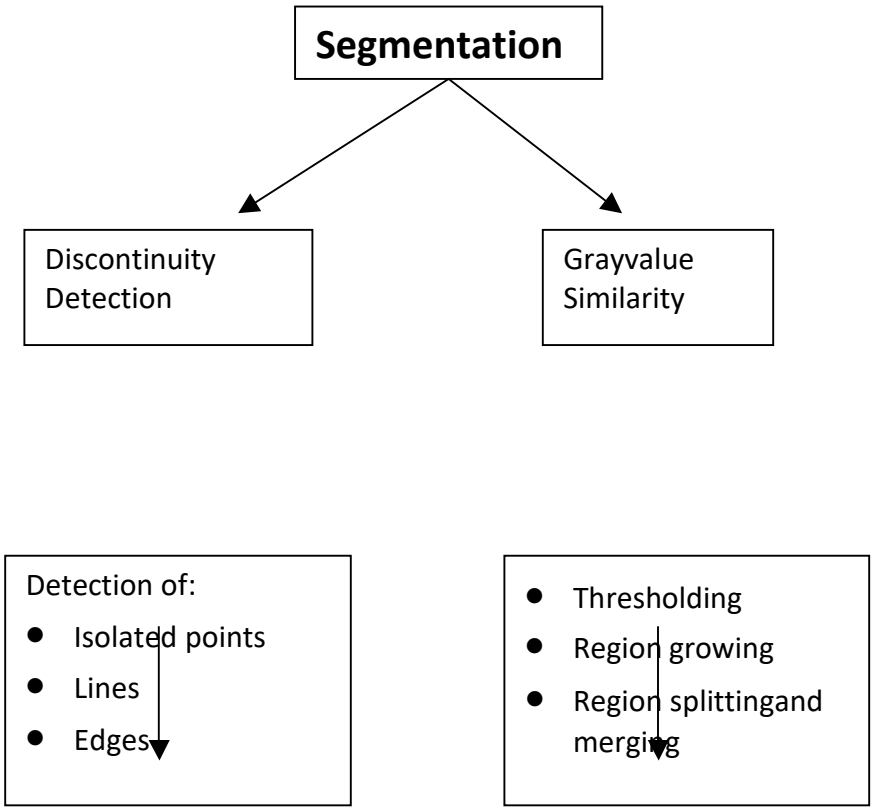




Given Image Segmented Image

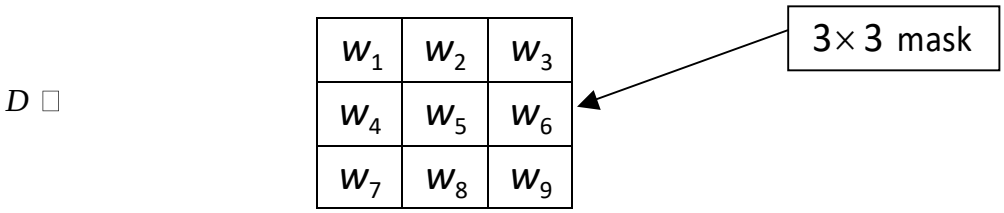
- It is an important and usually difficult step in image processing.

- Accurate segmentation determines the accuracy of subsequentimage analysis steps.


Segmentation
Discontinuity Detection
Grayvalue Similarity
Detection of:
Isolated points
Lines
Edges
Thresholding
Region growing
Region splittingand merging

Detection of Discontinuities

- This is usually accomplished by applying a suitable mask to theimage.



D □

f_1 f_2 f_3
f_4 f_5 f_6 □ R
f_7 f_8 f_9

Unit-5 (Image Segmentation & Representation)

Edge Detection

Lecture-3

Edge Based Segmentation

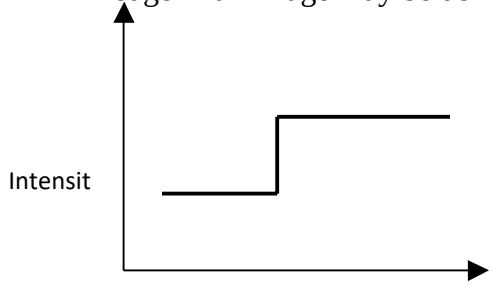
Edge detection is the process of locating edges in an image which is a very important step towards understanding image features. It is believed that edges consist of meaningful features and contains significant information. It significantly reduces the size of the image that will be processed and filters out information that may be regarded as less relevant, preserving and focusing solely on the important structural properties of an image for a business problem.

Edge-based segmentation algorithms work to detect edges in an image, based on various discontinuities in grey level, colour, texture, brightness, saturation, contrast etc. To further enhance the results, supplementary processing steps must follow to concatenate all the edges into edge chains that correspond better with borders in the image.

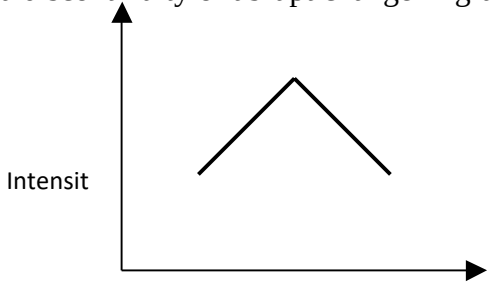
Edge detection algorithms fall primarily into two categories – Gradient based methods and Gray Histograms. Basic edge detection operators like sobel operator, canny, Robert's variable etc are used in these algorithms. These operators aid in detecting the edge discontinuities and hence mark the edge boundaries. The end goal is to reach at least a partial segmentation using this process, where we group all the local edges into a new binary image where only edge chains that match the required existing objects or image parts are present.

Edge Detection

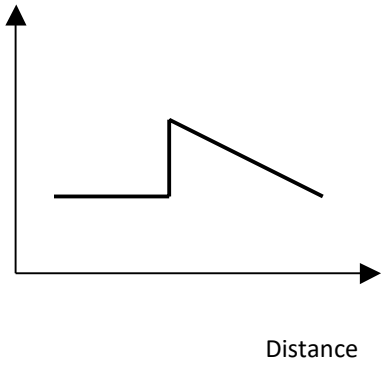
- Isolated points and thin lines do not occur frequently in mostpractical applications.
- For image segmentation, we are mostly interested in detecting the boundary between two regions with relatively distinct gray-level properties.
- We assume that the regions in question are sufficiently homogeneous so that the transition between two regions can be determined on the basis of gray-level discontinuities alone.
- An edge in an image may be defined as a discontinuity or abrupt change in gray level.



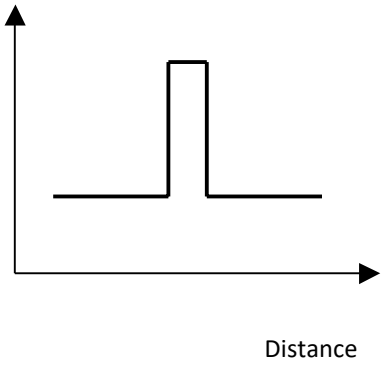
Ideal Step Edge



Ideal Roof Edge



Combination of Stepand Roof Edges

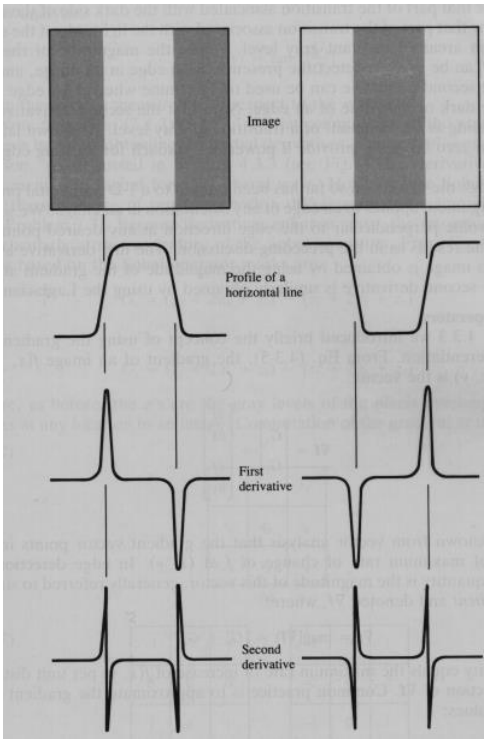


Ideal Spike Edge

- These are ideal situations that do not frequently occur in practice. Also, in two dimensions edges may occur at any orientation.

- Edges may not be represented by perfect discontinuities. Therefore, the task of edge detection is much more difficult than what it looks like.

- A useful mathematical tool for developing edge detectors is the first and second **derivative operator**s.



- From the example above it is clear that the **magnitude** of the **first derivative** can be used to detect the presence of an edge in an image.

- The **sign** of the **second derivative** can be used to determine whether an edge pixel lies on the dark or light side of an edge.

- The **zero crossings** of the **second derivative** provide a powerful way of locating edges in an image.

- We would like to have small-sized masks in order to detect fine variation in graylevel distribution (i.e., micro-edges).
- On the other hand, we would like to employ large-sized masks in order to detect coarse variation in graylevel distribution (i.e., macro-edges) and filter-out noise and other irregularities.
- We therefore need to find a mask size, which is a compromise between these two opposing requirements, or determine edge content by using different mask sizes.
- Most common differentiation operator is the gradient.

$$\nabla f(x,y) = \begin{bmatrix} \frac{\partial f(x,y)}{\partial x} \\ \frac{\partial f(x,y)}{\partial y} \end{bmatrix}$$

- The magnitude of the gradient is:

$$|\nabla f(x,y)| = \left[\left(\frac{\partial f(x,y)}{\partial x} \right)^2 + \left(\frac{\partial f(x,y)}{\partial y} \right)^2 \right]^{1/2}$$

- The direction of the gradient is given by :

$$\angle \nabla f(x,y) = \tan^{-1} \left\{ \frac{\partial f}{\partial y} \Big/ \frac{\partial f}{\partial x} \right\}$$

- In practice, we use discrete approximations of the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, which are implemented using the masks:

Unit-5 (Image Segmentation & Representation)

Thresholding

Lecture-4

Image Segmentation

It is a process of subdividing an image into the constituent parts or objects in the image.

The main purpose of subdividing an image into its constituent parts or objects present in the image is that to further analyze each of these constituents or each of the objects present in the image once they are identified.

Edge-Based Segmentation

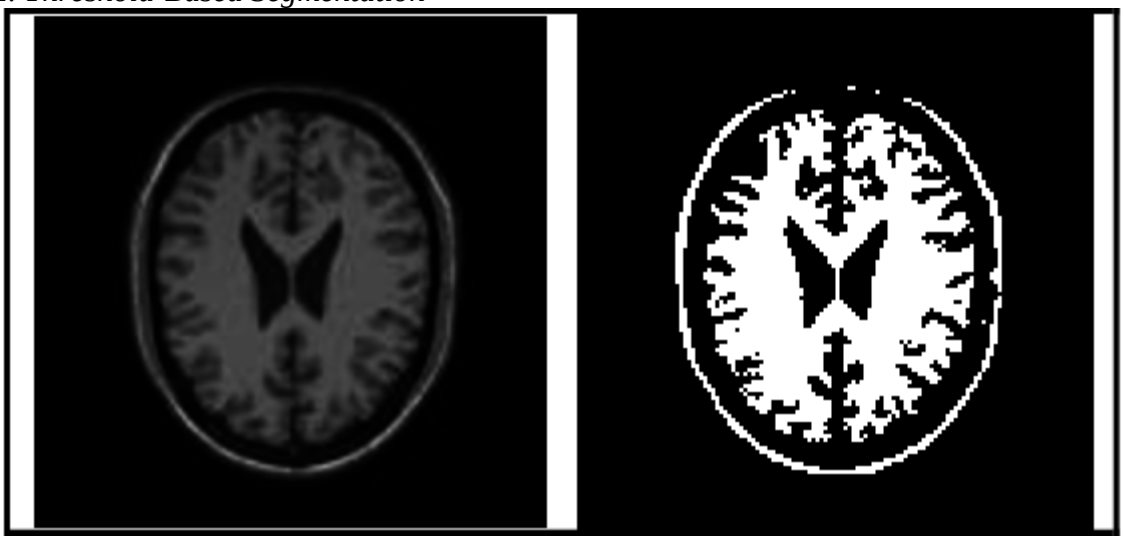


Source: ResearchGate

Edge-based segmentation is a popular image processing technique that identifies the edges of various objects in a given image. It helps locate features of associated objects in the image using the information from the edges. Edge detection helps strip images of redundant information, reducing their size and facilitating analysis.

Edge-based segmentation algorithms identify edges based on contrast, texture, color, and saturation variations. They can accurately represent the borders of objects in an image using edge chains comprising the individual edges.

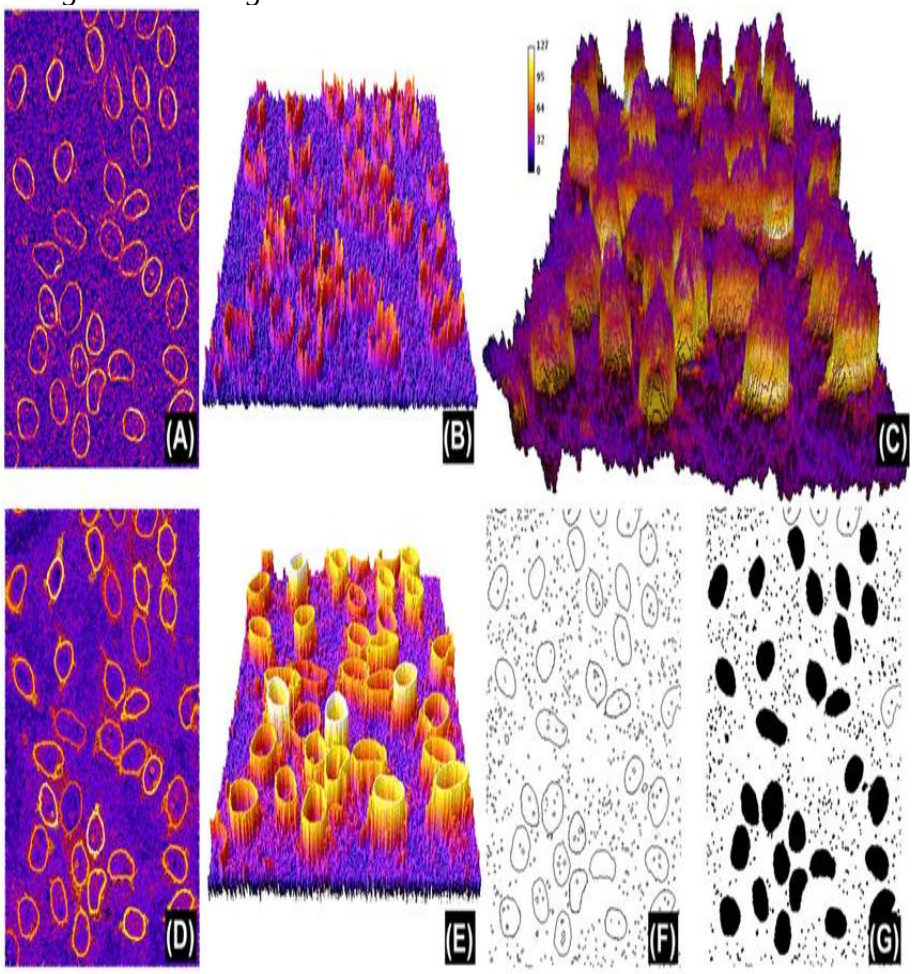
2. Threshold-Based Segmentation



Thresholding is the simplest image segmentation method, dividing pixels based on their intensity relative to a given value or threshold. It is suitable for segmenting objects with higher intensity than other objects or backgrounds.

The threshold value T can work as a constant in low-noise images. In some cases, it is possible to use dynamic thresholds. Thresholding divides a grayscale image into two segments based on their relationship to T, producing a binary image.

3. Region-Based Segmentation



Region-based segmentation involves dividing an image into regions with similar characteristics. Each region is a group of pixels, which the algorithm locates via a seed point. Once the algorithm

finds the seed points, it can grow regions by adding more pixels or shrinking and merging them with other points.

Unit-5 (Image Segmentation & Representation)

Edge and Boundary Linking

Lecture-5

There are three basic types of discontinuities in a digital image: Points, lines and edges. In practice, the most common way to look for discontinuities is to run a mask through the image. For the 3 x 3 mask ., this procedure involves computing the sum of products of the coefficients with the gray levels contained in the region encompassed by the mask. That is, the response of the mask at any point in the image

Where is the gray level of the pixel associated with mask coefficient As Wi. usual, the response of the mask is defined with respect to its center location. When the mask is centered on a boundary pixel, the response is computed by using the appropriated partial neighbor hood.

Point Detection

The detection of isolated points in an image is straightforward. we say that a point has been detected at the location on which the mask is centered if

Where T is a nonnegative threshold and R is the response of the mask at any point in the image. Basically all that this formulation does is measure the weighted differences between the center point and its neighbors. The idea is that the gray level of an isolated point will be quite different from the gray level of its neighbors.

Edge Detection

The most common approach for detecting meaningful discontinuities in gray level. We discuss approaches for implementing first-order derivative (Gradient operator), second-order derivative

(Laplacian operator).

Basic Formulation

An edge is a set of connected pixels that lie on the boundary between two regions. An edge is a "local" concept whereas a region boundary, owing to the way it is defined, is a more global idea.

We start by modeling an edge intuitively. This will lead us to formalism in which "meaningful" transitions in gray levels can be measured.

In practice, optics, sampling, and other acquisition imperfections yield edges that are blurred, with the degree of blurring being determined by factors such as the quality of the image acquisition system, the sampling rate, and illumination conditions under which the image is acquired.

The slope of the ramp is inversely proportional to the degree of blurring in the edge. In this model, we no longer have a thin (one pixel thick) path. Instead, an edge point now is any point contained in the ramp, and an edge would then be a set of such points that are connected. The thickness is determined by the length of the ramp. The length is determined by the slope, which is in turn determined by the degree of blurring. Blurred edges tend to be thick and sharp; edges tend to be thin.

The first derivative is positive at the points of transition into and out of the ramp as we move from left to right along the profile; it is constant for points in the ramp; and is zero in areas of constant gray level. The second derivative is positive at the transition associated with the light side of the edge, and zero along the ramp and in areas of constant gray level. The following are the two additional properties of the second derivative around an edge: It produces two values for every edge in an image (an undesirable feature) An imaginary straight line joining the extreme positive and negative values of the second derivative would cross zero near the midpoint of the edge (zero-crossing property)

B) Edge Linking and Boundary Detection

Edge linking and boundary detection operations are the fundamental steps in any image understanding. Edge linking process takes an unordered set of edge pixels produced by an edge detector as an input to form an ordered list of edges. Local edge information are utilized by edge linking operation; thus edge detection algorithms typically are followed by linking procedure to assemble edge pixels into meaningful edges.

9.3.1 Local Processing

One of the simplest approaches of linking edge points is to analyze the characteristics of pixels in a small neighborhood (say, 3 x 3 or 5 x 5) about every point (x, y) in an image that has undergone edge-detection. All points that are similar are linked, forming a boundary of pixels that share some common properties.

The two principal properties used for establishing similarity of edge pixels in this kind of analysis are (1) the strength of the response of the gradient operator used to produce the edge pixel; and (2) the direction of the gradient vector. The first property is given by the value of , the gradient. Thus an edge pixel with coordinates in a predefined neighborhood of (x, y) is similar in magnitude to the pixel at (x, y) if

where E is a non negative threshold

The direction (angle) of the gradient vector is given by. An edge pixel at in the predefined neighborhood of (x, y) has an angle similar to the pixel at (x, y) if

where A is a nonnegative angle threshold. As noted in 9.2.3, the direction of the edge at (x, y) is *perpendicular* to the direction of the gradient vector at that point.

A point in the predefined neighborhood of (x, y) is linked to the pixel at (x, y) if both magnitude and direction criteria are satisfied. This process is repeated at every location in the image. A record must be kept of linked points as the center of the neighborhood is moved from pixel to pixel. A simple book keeping procedure is to assign a different gray level to each set of linked edge pixels.

9.3.2 Global Processing via the Hough transformIn this section, points are linked by determining first if they lie on a curve of specified shape. Unlike the local analysis method, we now consider global relationships between pixels.

Suppose that, for n points in an image, we want to find subsets of these points that lie on straight lines. One possible solution is to first find all lines determined by every pair of points and then find all subsets of points that are close to particular lines. The problem with this procedure is that it involves finding n(n-1)/2 lines and then performing (n)(n(n-1))/2 comparisons of every point to all lines. This approach is computationally prohibitive in al but the most trivial applications.

9.3.3 Global Processing via Graph-Theoretic Techniques

In this section, a global approach based on representing edge segments in the form of a graph and searching the graph for low-cost paths that correspond to significant edges is discussed. This representation provides a rugged approach that performs well in the presence of noise. As might be expected, the procedure is considerably more complicated and requires more processing time.

A graph G = (N, A) is a finite, non empty set of nodes N, together with a set A of unordered pair of distinct elements of N. Each pair of A is called an arc. A graph in which the arcs are directed is called a directed graph. If an arc is directed from node to node, then is said to be a successor of its parent node . The process of identifying the successors of a node is called expansion of the node. In each graph we define levels, such that level consists of a single node, called the start node, and the nodes in the last level are called goal nodes. A cost can be associated with every arc . A sequence of nodes with each being a successor of node is called a path from , and the cost of the path is

Unit-5 (Image Segmentation & Representation)

Hough Transforms

Lecture-6

- The Hough transform (HT) can be used to detect lines, circles or other parametric curves
- It was introduced in 1962 (Hough 1962) and first used to find lines in images a decade later (Duda 1972).
- The goal is to find the location of lines in images.
- This problem could be solved by e g Morphology and a linear â€¢ This problem could be solved by e.g. Morphology and a linear structuring element, or by correlation.
- Hough transform can detect lines, circles and other structures if their parametric equation is known.
- It can give robust detection under noise and partial occlusion â€¢ It can give robust detection under noise and partial occlusion.

The Hough transform (HT) is a feature extraction approach in image analysis, computer vision, and digital image processing [1]. It uses a voting mechanism to identify bad examples of objects inside a given class of forms. This voting mechanism is carried out in parameter space. Object candidates are produced as local maxima in an accumulator space using the HT algorithm.

The HT is a method for separating features of a certain shape inside a picture. The classical HT is most typically employed to detect regular curves such as lines, circles, ellipses, etc. It needs the required features to be provided in some parametric form. A generalized HT can be used when a brief analytic description of a feature(s) is not attainable. We will limit our study to the traditional HT due to the computational difficulty of the extended Hough algorithm. Despite its domain limitations, the classical HT has many applications. Most manufactured parts (as well as many anatomical parts investigated in medical imagery) have feature boundaries that standard curves can describe. The real benefit of the HT approach is that it is unaffected by picture noise and is tolerant of gaps in feature boundary descriptions.

This article will provide you with in-depth knowledge about Hough transform. It will give you a basic introduction to hough transform how exactly it works and what is the maths behind it. It also enlists merits and demerits of hough transforms with its various applications.

History

It was first developed for the automated examination of bubble chamber pictures (Hough, 1959). In 1962, the Hough transform was patented as U.S. Patent 3069654 and assigned Atomic Energy Commission to the United States as a “Method and Means for Recognizing Complex Patterns.” Because the slope might go to infinity, this invention employs a slope-intercept parameterization for straight lines, which results in infinite transform space.

Duda, R. O., and P. E. Hart, “Use of the HT to Detect Lines and Curves in Pictures,” Comm. ACM, Vol. 15, pp. 11–15 (January 1972), although it had been standard for the Radon transformation since the 1930s. In Frank O’Gorman, MB Clowes: Finding Picture Edges through Collinearity of Feature Points, O’Gorman and Clowes discuss their version. IEEE Transactions on Computers, vol. 25, no. 4, pp. 449-456. (1976) Hart, P. E., “How the HT was invented,” IEEE Signal Processing Magazine, Vol 26, Issue 6, pages 18 – 22 (November 2009) tells the tale of how the contemporary form of the HT was invented.

What is Hough Transform?

The HT is a feature extraction method in image analysis, computer vision, and digital image processing [3]. It uses a voting mechanism to identify bad examples of objects inside a given class of forms. This voting mechanism is carried out in parameter space. First, object candidates are produced as local maxima in an accumulator space, using the HT algorithm.
The traditional HT was concerned with detecting lines in an image, but it was subsequently expanded to identifying locations of arbitrary shapes, most often circles or ellipses. Richard Duda and Peter Hart devised the HT as we knew it today in 1972, calling it a “generalized HT”[4] after Paul Hough’s related 1962 patent[5]. Dana H. Ballard popularized the transformation in the computer vision field with his 1981 journal paper “Generalizing the HT to identify arbitrary forms.”

Why is it Needed?

In many circumstances, an edge detector can be used as a pre-processing stage to get picture points or pixels on the required curve in the image space. However, there may be missing points or pixels on the required curves due to flaws in either the image data or the edge detector and spatial variations between the ideal line/circle/ellipse and the noisy edge points acquired by the edge detector. As a result, grouping the extracted edge characteristics into an appropriate collection of lines, circles, or ellipses is frequently difficult.



1. Original image of Lane

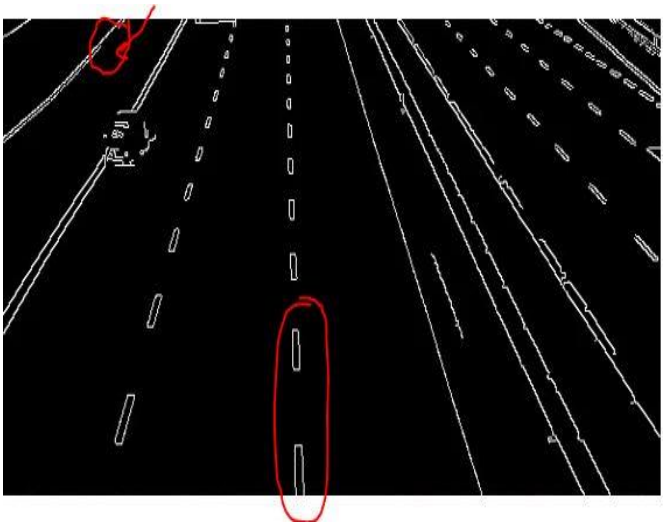
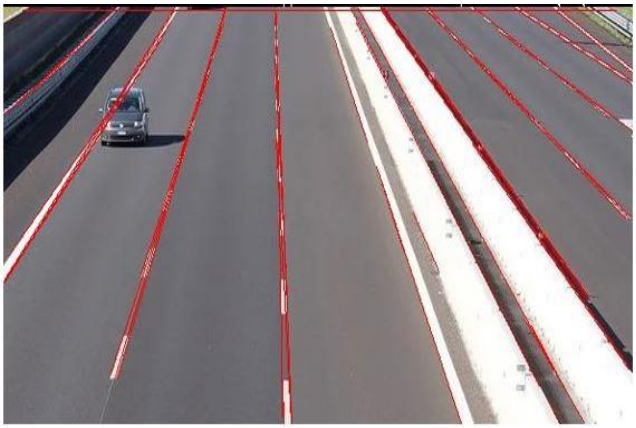


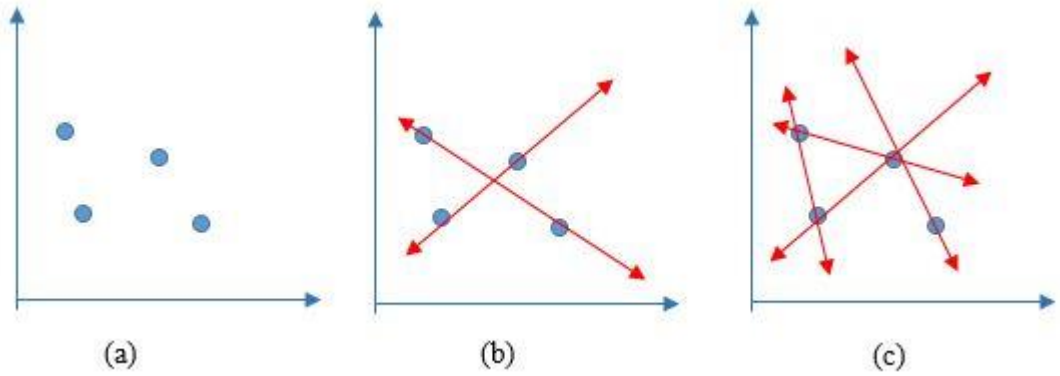
Figure 2: Image after applying edge detection technique. Red circles show that the line is breaking there.



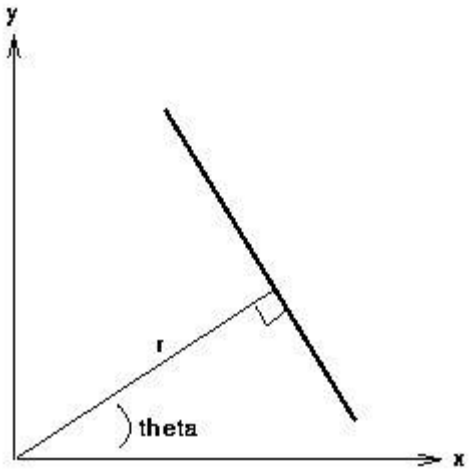
How Does it Work?

The Hough approach is effective for computing a global description of a feature(s) from (potentially noisy) where the number of solution classes does not need to be provided before. For example, the Hough approach for line identification is motivated by the assumption that each input

measurement reflects its contribution to a globally consistent solution (e.g., the physical line which gave rise to that image point).



A line can be described analytically in a variety of ways. One of the line equations uses the parametric or normal notion: $x\cos\theta+y\sin\theta=r$. where r is the length of a normal from the origin to this line and θ is the orientation as given in Figure 5.



The known variables (i.e., x_i,y_i) in the Image are constants in the parametric line equation, whereas r and are the unknown variables we seek. Points in cartesian image space correspond to curves (i.e., sinusoids) in the polar Hough parameter space if we plot the potential (r, θ) values specified by each. The Hough transformation for straight lines is this point-to-curve transformation. Collinear spots in the cartesian image space become clearly obvious when examined in the Hough parameter space because they provide curves that overlap at a single (r, θ) point.

Where a and b are the circle's center coordinates, and r is the radius. Because we now have three coordinates in the

parameter space and a 3-D accumulator, the algorithm's computing complexity increases. (In general, the number of parameters increases the calculation and the size of the accumulator array polynomially.) As a result, the fundamental Hough approach described here is only applicable to straight lines.

Algorithm

1. Determine the range of ρ and θ. Typically, the range of θ is [0, 180] degrees and ρ is [-d, d], where d is the diagonal length of the edge. Therefore, it's crucial to quantify the range of ρ and θ, which means there should only be a finite number of potential values.
2. Create a 2D array called the accumulator with the dimensions (num rhos, num thetas) to represent the Hough Space and set all its values to zero.
3. Use the original Image to do edge detection (ED). You can do this with whatever ED technique you like.
4. Check each pixel on the edge picture to see if it is an edge pixel. If the pixel is on edge, loop over all possible values of θ, compute the corresponding ρ, locate the θ and ρ index in the accumulator, then increase the accumulator base on those index pairs.
5. Iterate over the accumulator's values. Retrieve the ρ and θ index, and get the value of ρ and θ from the index pair, which may then be transformed back to the form of y = ax + b if the value is greater than a specified threshold.

Sum of Hough Transform

Problem: Given set of points, use Hough transform to join these points. A(1,4), B(2,3) ,C(3,1) ,D(4,1) ,E(5,0)

Solution:

Lets think about equation of line that is y=ax+b.

Now, if we rewrite the same line equation by keeping b in LHS, then we get

b=-ax+y. So if we write the same equation for point A(1,4), then consider x=1 and y=4 so that we will get

b=-a+4. The following table shows all the equations for a given point

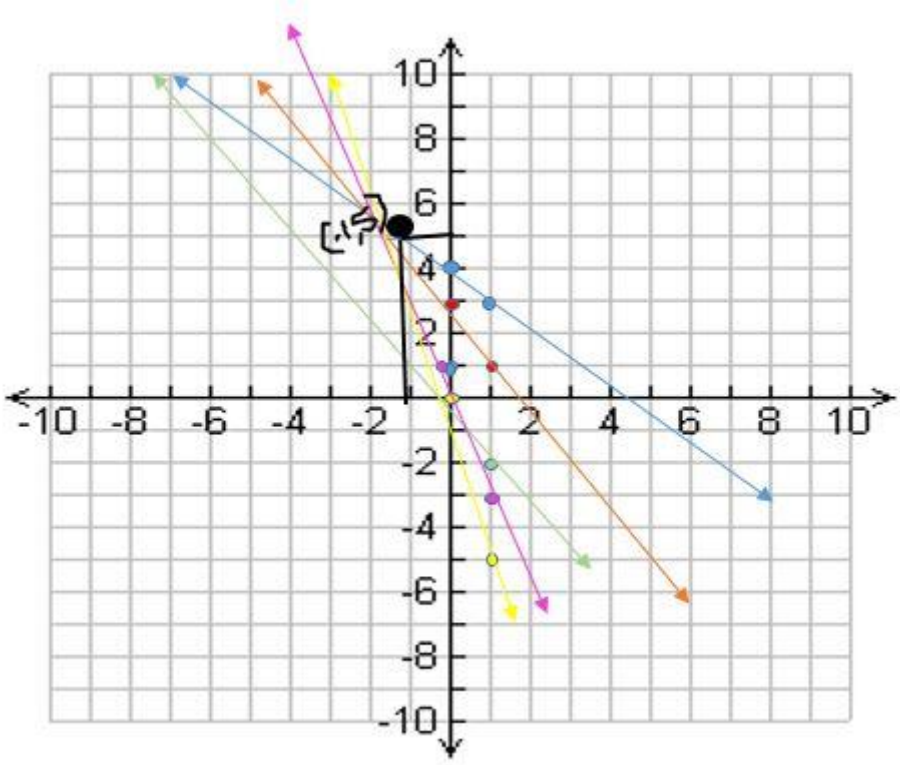
Point	X and y values	Substituting the value in b=-ax+y
A(1,4)	x=1 ; y=4	b= -a+4
B(2,3)	x=2 ; y=3	b= -2a+3
C(3,1)	x=3 ; y=1	b= -3a+1

D(4,1)	x=4 ; y=1	b= -4a+1
E(5,0)	x=5 ; y=0	b= -5a+0

Now take x-0 and find corresponding y value for above given five equations

Point	equations	Now a=0	New point (a,b)	Now a=1
A(1,4)	b= -a+4	b= -(0)+4 =4	(0,4)	b= -(1)+4 =
B(2,3)	b= -2a+3	b= -2(0)+3=3	(0,3)	b= -2(1)+3
C(3,1)	b= -3a+1	b= -3(0)+1=1	(0,1)	b= -3(1)+1
D(4,1)	b= -4a+1	b= -4(0)+1=1	(0,1)	b= -4(1)+1
E(5,0)	b= -5a+0	b= -5(0)+0=0	(0,0)	b= -5(1)+0

Let us plot the new point on the graph as given below in figure 6.



We can see that almost all line crosses each other at a point (-1,5). So here now a=-1 and b =5.

Now let's put these values in the y=ax+b equation so we get y=-1x+5 so y=-x+5 is the line equation that will link all the edges.

Advantages

The HT benefits from not requiring all pixels on a single line to be contiguous. As a result, it can be quite effective when identifying lines with small gaps due to noise or when objects are partially occluded.

Application

The HT has been widely employed in numerous applications because of the benefits, such as noise immunity. 3D applications, object and form detection, lane and traffic sign recognition, industrial

and medical applications, pipe and cable inspection, and underwater tracking are just a few examples [6]. Below are some examples of these applications. [7] Proposes the hierarchical additive Hough transform (HAHT) for detecting lane lines. The HAHT that is recommended accumulates the votes at various hierarchical levels. Line segmentation into multiple blocks also minimizes the computational load. [7] proposes a lane detection strategy in which the HT is merged with the joint photographic experts' group (JPEG) compression. However, only simulations are used to test the method.

Biometric and Man-machine Interaction

A generalized hand detection model with articulated degrees of freedom is shown in [6]. It makes use of geometric feature correspondences between points and lines. The lines are recognized by the PHT. Following that, they are matched with a model. The GHT can identify Turkish finger-spelling in [6]. To produce interest zones, the scale-invariant features transform is applied (SIFT). The superfluous interest zones are eliminated using skin color reduction. [6] also suggests using hand gestures and tracking to create a real-time human-robot interaction system. In this procedure, the linked component labeling (CCL) and the HT are integrated. By sensing the skin tone, it can extract the hand's center, orientations, and fingertip positions of all outstretched fingers.

3D Applications

A detector for 3D lines, boxes, and blocks is proposed in [6]. This approach decreases the computing cost by using 2D Hough space for parallel line detection. In addition, the 3D HT is used to extract planar faces from point clouds that are unevenly dispersed.

Object Recognition

It describes a method for detecting objects utilizing the GHT and color similarity between homogeneous portions of the item. It takes as input already segmented areas of homogenous hue. According to this research, it is resistant to changes in light, occlusion, and distortion of the segmentation output. It can distinguish items that have been rotated, scaled, and even moved around in a complicated environment.

Object Tracking

It uses a blend of uniform and Gaussian Hough transforms to conduct shape-based tracking. While the camera moves, this method can find and follow objects even against difficult backdrops like a dense forest.

Underwater Application

It provides a technique for detecting geometrical forms in underwater robot images. The

recognition problem is transformed into a bounded error estimation problem, as opposed to the traditional GHT. On an autonomous underwater vehicle, [6] creates an underwater system for visually directed operations (AUV). Underwater pipelines are identified using the HT. Their orientations are then determined using binary morphology.

Industrial and Commercial Application

The HT and its various versions have various industrial and commercial uses. In, an uncrewed aerial vehicle (UAV) surveillance and inspection system are used to propose a knowledge-based power line identification approach. Before applying the LHT, a pulse-coupled neural network (PCNN) filter is constructed to eliminate background noise from picture frames. The findings are refined using knowledge-based line clustering.

Medical Application

It uses a warped frequency transform (WFT) to adjust for the dispersive behavior of ultrasonic guided waves, followed by a Wigner-Ville time-frequency analysis and the HT to increase fault localization accuracy. The locations and boundaries of vertebrae are recognized automatically. The HT is used in conjunction with the GA to determine the migrating vertebrae.

Unconventional Application

This section shows unexpected usage to demonstrate the HT's adaptability and ever-expanding nature. It proposes a data mining technique for locating arbitrarily oriented subspace clusters. This is done by mapping the data space to the parameter space, which defines the collection of arbitrarily oriented subspaces that may be created. The clustering method works by locating clusters that may hold many database objects. Even if subspace clusters of differing dimensions are sparse or crossed by other clusters in a noisy environment, the method can discover them.

Unit-5 (Image Segmentation & Representation)

Region Based Segmentation

Lecture-7

Region and Edge Based Segmentation

Segmentation

Segmentation is the separation of one or more regions or objects in an image based on a discontinuity or a similarity criterion. A region in an image can be defined by its border (edge) or its interior, and the two representations are equal. There are prominently three methods of performing segmentation:

- Pixel Based Segmentation
- Region-Based Segmentation
- Edges based segmentation

Edges based segmentation

Edge-based segmentation contains 2 steps:

- **Edge Detection:** In edge detection, we need to find the pixels that are edge pixels of an object. There are many object detection methods such as Sobel operator, Laplace operator, Canny, etc.

1	**0**	**-1**
2	**0**	**-2**
1	**0**	**-1**

Sobel vertical Operator

+1	**2**	**1**
0	**0**	**0**
-1	**-2**	**-1**

Sobel Horizontal Operator

0	**-1**	**0**
-1	**4**	**-1**
0	**-1**	**0**

Negative Laplace Operator

- **Edge Linking:** In this step, we try to refine the edge detection by linking the adjacent edges and combine to form the whole object. The edge linking can be performed using any of the two methods below:
    - Local Processing: In this method, we used gradient and direction to link the neighborhood edges. If two edges have a similar direction vector then they can be linked.
    - Global processing: This method can be done using HOG transformation
- **Pros** :
    - This approach is similar to how the humans brain approaches the segmentation task.
    - Works well in images with good contrast between object and background.
- **Limitations:**
    - Does not work well on images with smooth transitions and low contrast.
    - Sensitive to noise.
    - Robust edge linking is not trivial and easy to perform.

Region-Based Segmentation

In this segmentation, we grow regions by recursively including the neighboring pixels that are similar and connected to the seed pixel. We use similarity measures such as differences in gray levels for regions with homogeneous gray levels. We use connectivity to prevent connecting different parts of the image.

There are two variants of region-based segmentation:

- **Top-down approach**
    - First, we need to define the predefined seed pixel. Either we can define all pixels as seed pixels or randomly chosen pixels. Grow regions until all pixels in the image belongs to the region.
- **Bottom-Up approach**
    - Select seed only from objects of interest. Grow regions only if the similarity criterion is fulfilled.
- **Similarity Measures:**
    - Similarity measures can be of different types: For the grayscale image the similarity measure can be the different textures and other spatial properties, intensity difference within a region or the distance b/w mean value of the region.
- **Region merging techniques:**
    - In the region merging technique, we try to combine the regions that contain the single object and separate it from the background.. There are many regions merging techniques such as Watershed algorithm, Split and merge algorithm, etc**.**
- **Pros:**
    - Since it performs simple threshold calculation, it is faster to perform.
    - Region-based segmentation works better when the object and background have high contrast.
- **Limitations:**
    - It did not produce many accurate segmentation results when there are no significant differences b/w pixel values of the object and the background.

Implementation:

- In this implementation, we will be performing edge and region-based segmentation. We will be using scikit image module for that and an image from its dataset provided.

Unit-5 (Image Segmentation & Representation)

Boundary Representation & Boundary Descriptors

Lecture-8

Boundary descriptors, such as boundary length, diameter, curvature, etc. - regional descriptors, such as area, perimeter, compactness, mean value, etc. Generally, an external representation is chosen when a primary focus is on shape characteristics.

Region representation:

- based on external characteristics (its boundary)
- based on internal characteristics (pixels comprising the region)

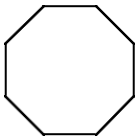
Region description:

- boundary descriptors, such as boundary length, diameter, curvature, etc.
- regional descriptors, such as area, perimeter, compactness, mean value, etc.

Generally, an external representation is chosen when a primary focus is on ***shape*** *characteristics. An internal representation is selected when a primary focus is on* ***reflectivity properties, such as color or texture.***

REPRESENTATION SCHEMES

The segmentation techniques provide results in form of pixels along the boundary or pixels contained in a region:





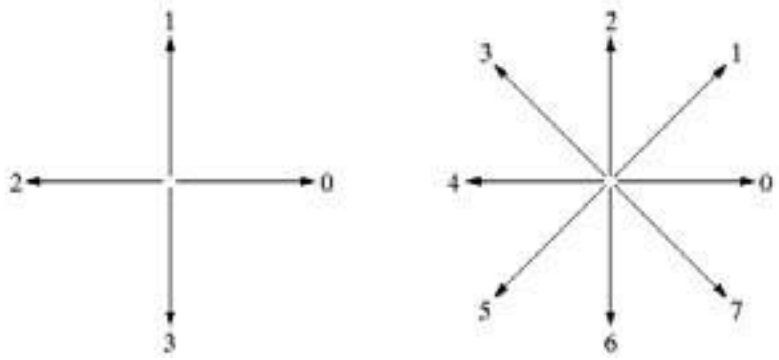
It is a standard practice to use boundary representation schemes that **compact** the data into representations that are more useful in the computation of descriptors. One of the schemes are chain codes.

Chain codes

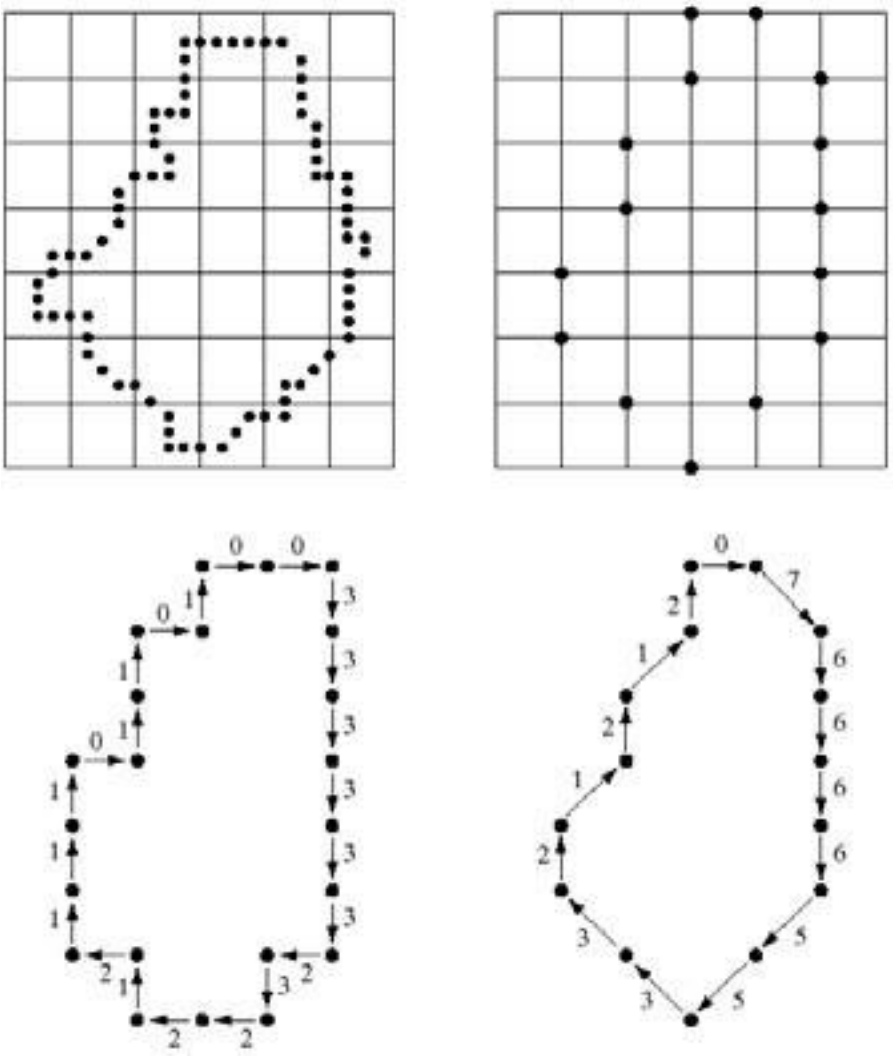
- represent a boundary by a connected sequence of straight-line segments of specified length and direction
- the direction of each segment is coded by using a numbering scheme:

a b

FIGURE 11.1 Direction numbers for (a) 4-directional chain code, and (b) 8-directional chain code.



- a typical chain code is generated by following a boundary in a clockwise direction and assigning a direction to the segments connecting every pair of pixels
  - this method results in quite long chain codes
  - any small disturbances along the boundary may cause changes in code that may not be related to the shape of the boundary
- another technique is to select a larger grid spacing, assign boundary points to each node of the large grid (depending on the proximity of the original boundary to that node), and represent the resampled boundary by a 4- 8-code:



a b
c d

FIGURE 11.2
(a) Digital boundary with resampling grid superimposed. (b) Result of resampling. (c) 4-directional chain code. (d) 8-directional chain code.

the boundary representation in figure (c) is: 003333332322121110110 1 the boundary representation in figure (d) is: 076666553321212

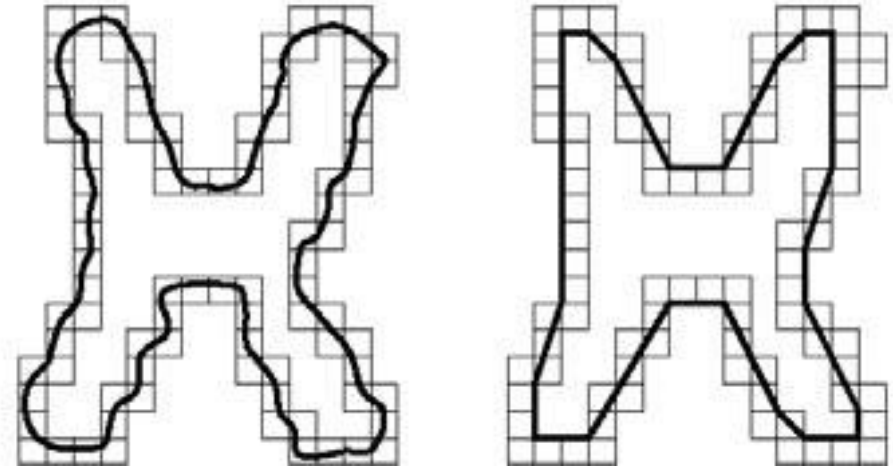
The accuracy of the resulting code depends on the spacing of the sampling grid.

Polygonal approximations

- a digital boundary can be approximated by a polygon
- **minimum perimeter** polygons
  - enclose a boundary by a set of concatenated cells and produce a minimum perimeter that fits the cell strip:

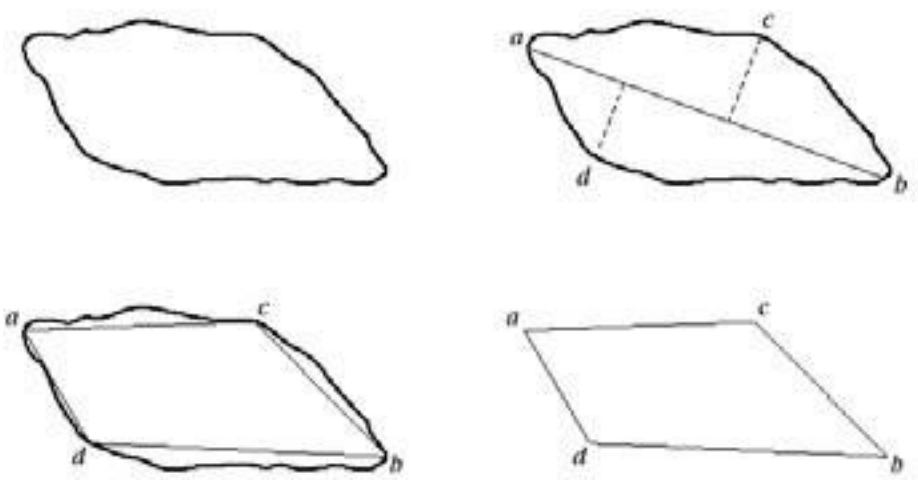
a b

FIGURE 11.3
(a) Object boundary enclosed by cells. (b) Minimum perimeter polygon.



- **merging** techniques (merge points along a boundary until the least square error line fit of the points merged exceeds a preset threshold; repeat the procedure for the new points along the boundary; at the end of the procedure the intersections of adjacent line segments form the vertices of the polygon)

- **splitting** techniques (subdivide a segment successively into two parts until a given criterion is satisfied -- for example, a requirement might be that the maximum perpendicular distance from a boundary segment to the line joining two end points does not exceed a preset threshold)



a b
c d

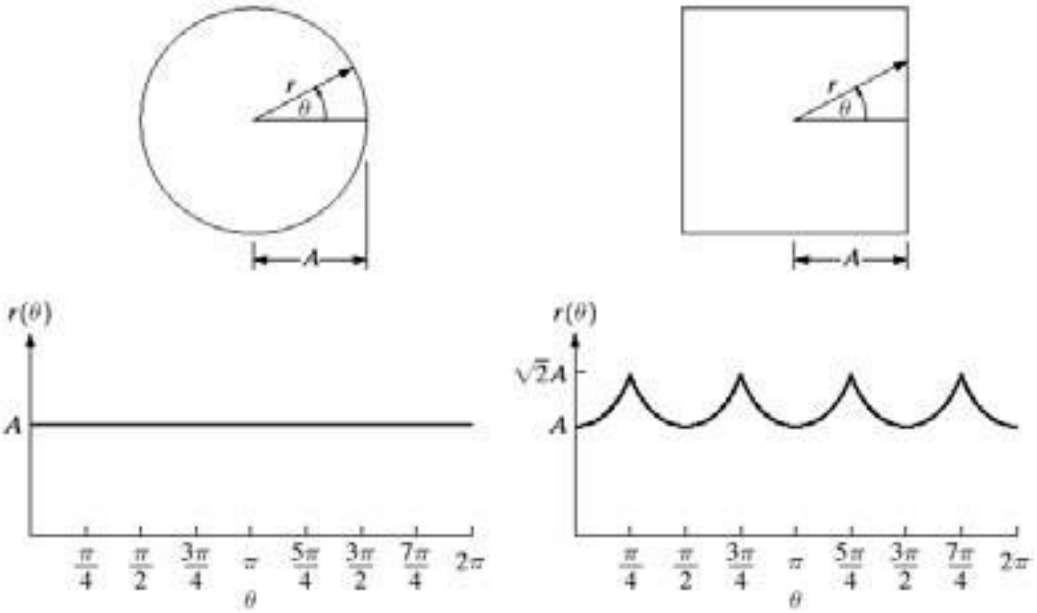
FIGURE 11.4
(a) Original boundary. (b) Boundary divided into segments based on extreme points. (c) Joining of vertices. (d) Resulting polygon.

Signatures

- a simple functional representation that can be used to describe and reconstructthe boundary with appropriate accuracy
- the simplest signature is a **plot of the distance from the centroid to theboundary as a function of angle**:

a b

FIGURE 11.5
Distance-versus-angle signatures. In (a) $r(\theta)$ is constant. In (b), the signature consists of repetitions of the pattern $r(\theta) = A\sec\theta$ for $0 \le \theta \le \pi/4$ and $r(\theta) = A\csc\theta$ for $\pi/4 < \theta \le \pi/2$.



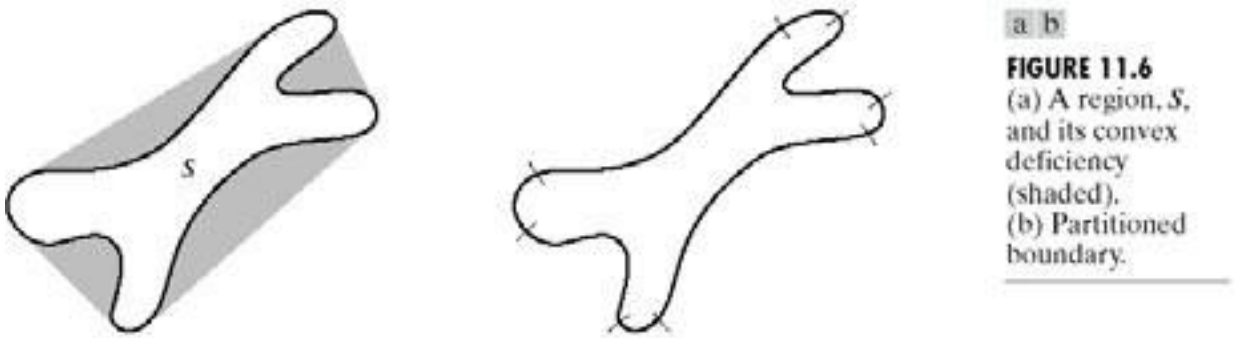
• signature is an translation-invariant representation which allows an object to be compared with a standard prototype by cyclically shifting the signature of one with respect to the other in steps, while checking for the best match

Boundary segments

• **convex hull** technique

- convex hull H of the arbitrary set S is the smallest convex set containing S; the set H-S is called convex deficiency D of the set S

- the region boundary can be partitioned by following the contour of S and marking the points at which a transition is made into or out of a componentof the convex deficiency



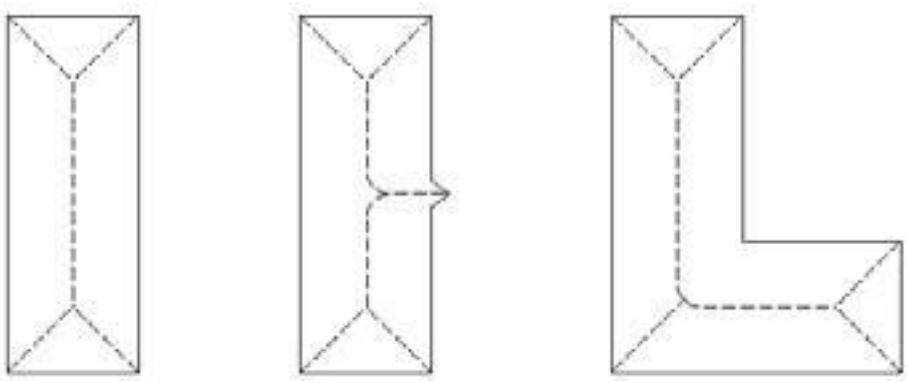
a b

FIGURE 11.6
(a) A region, *S*, and its convex deficiency (shaded). (b) Partitioned boundary.

- in practice, the boundary is usually smoothed prior to partitioning (for example, by using a polygonal approximation)

The skeleton of a region

- **medial axis transformation** (MAT) generates a "skeleton" of a region
- MAT algorithm:
  (1) for each point in the region we find its closest point in boundary,
  (2) if a point has more than one such a neighbor --> a point belongs tothe medial axis (skeleton) of the region



a b c
FIGURE 11.7
Medial axes (dashed) of three simple regions.

- the results of MAT operation depend on the **distance measure**: pixel

coordinates

p (x,y)
q (s,t)

Euclidean distance between p and q is defined as:

$D_e(p,q)=[(x-s)^2 + (y-t)^2]^{1/2}$

D4 distance:

$D_4(p,q)=|x-s|+ |y-t|$

For example, the pixels with D4 distance □ 2 from pixel (x,y) form thefollowing contours of constant distance:

2
2 1 2
2 1 0 1 2
2 1 2
2

D8 distance:

D8(p,q)=max (|x-s|, |y-t|)

For example, the pixels with D8 distance □ 2 from pixel (x,y) form the following contours of constant distance:

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

The Laplacian Edge Detector

- In many applications, it is of particular interest to construct derivative operators, which are **isotropic** (rotation invariant). This means that rotating the image f and applying the operator gives the same result as applying the operator on f and then rotating the result.
- We want the operator to be isotropic because we want to equally sharpen edges which run in any direction.
- This can be accomplished by another detector, which is based on the **Laplacian** of a two-dimensional function.
- The Laplacian of a two-dimensional function $f(x, y)$ is given by

$$\nabla^2 f(x,y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- The Laplacian is rotation invariant!
- A discrete approximation can be obtained as:

$$\nabla^2 f(m,n) \approx 4f(m,n) - (f(m-1,n) + f(m+1,n) + f(m,n-1 + f(m,n+1))$$

- This can be implemented using the following mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

- Other approximations of the Laplacian can be obtained by means of the masks: